Polynomials BOS

3unit

NAME:

1	HSC 00		

Find the value of k if $x-3$ is a factor of $P(x)=x^3-3kx+6$. **2**

If $x-3$ is a factor of $P(x)$, then $P(3)=0$

i.e. $(3)^3-3k(3)+6=0$

$27-9k+6=0$

$33=9k$

$\boxed{k=\frac{33}{9}=\frac{11}{3}.}$

2	HSC 00		

The polynomial $P(x)=x^3+px^2+qx+r$ has real roots $\sqrt{k}, -\sqrt{k}$ and α. **4**

(i) Explain why $\alpha+p=0$.

(ii) Show that $k\alpha=r$.

(iii) Show that $pq=r$.

(i) The polynomial $P(x)=x^3+px^2+qx+r$ has roots:

$x_1=\sqrt{k}$, $x_2=-\sqrt{k}$ and $x_3=\alpha$

Now $x_1+x_2+x_3=-p$ | i.e. $\sqrt{k}-\sqrt{k}+\alpha=-p$

$\therefore \alpha+p=0$

(ii) Now $x_1x_2x_3=-r$

i.e. $\sqrt{k}(-\sqrt{k})\alpha=-r$

$\therefore -k\alpha=-r$

$\therefore k\alpha=r$

(iii) Now $x_1x_2+x_1x_3+x_2x_3=q$

$\therefore \sqrt{k}(-\sqrt{k})+\sqrt{k}\,\alpha+(-\sqrt{k})\alpha=q$

$\therefore -k+\sqrt{k}\,\alpha-\sqrt{k}\,\alpha=q$

$\therefore -k=q$

Also, $p=-\alpha$ (from (i))

$\therefore pq=-\alpha(-k)$

$=\alpha k$

$=r$ (from (ii))

3	HSC 01		

Is $x+3$ a factor of $x^3-5x+12$? Give reasons for your answer. **2**

Let $P(x)=x^3-5x+12$

$P(-3)=(-3)^3-5(-3)+12$

$=-27+15+12$

$=0$

$\therefore x+3$ is a factor of $x^3-5x+12$.

4	HSC 02		

Suppose $x^3-2x^2+a\equiv(x+2)Q(x)+3$ where $Q(x)$ is a polynomial. **2**

Find the value of a.

$x^3-2x^2+a\equiv(x+2)Q(x)+3$

For $x=-2$,

$(-2)^3-2(-2)^2+a=0\times Q(-2)+3$

$-8-8+a=3$

$\therefore a=19.$

5	HSC 02		

The polynomial $P(x)=x^3-2x^2+kx+24$ has roots α, β, γ.

(i) Find the value of $\alpha+\beta+\gamma$. **1**

(ii) Find the value of $\alpha\beta\gamma$. **1**

(iii) It is known that two of the roots are equal in magnitude but opposite in sign. Find the third root and hence find the value of k. **2**

(i) $\alpha+\beta+\gamma=2$ (ii) $\alpha\beta\gamma=-24$

(iii) Let $\alpha=-\beta$, then the roots are $-\beta$, β and γ.

$\therefore -\beta+\beta+\gamma=2$ (from (i))

$\therefore \gamma=2$

Now $k=\alpha\beta+\alpha\gamma+\beta\gamma$

$=(-\beta)\beta+(-\beta)\gamma+\beta\gamma$

$=-\beta^2-\beta\gamma+\beta\gamma$

$=-\beta^2$

But $\alpha\beta\gamma=-24$ (from (ii))

$\therefore (-\beta)\beta\gamma=-24$

$\therefore -\beta^2\gamma=-24$

$\therefore -2\beta^2=-24$ (since $\gamma=2$)

$\therefore -\beta^2=-12$

$\therefore k=-12.$ (since $k=-\beta^2$)

6	HSC 03		

It is known that two of the roots of the equation $2x^3+x^2-kx+6=0$ are reciprocals of each other. Find the value of k. **2**

$2x^3+x^2-kx+6=0$ . . . (1)

Now $\alpha+\frac{1}{\alpha}+\beta=-\frac{1}{2}$

and $\alpha\left(\frac{1}{\alpha}\right)+\alpha(\beta)+\frac{1}{\alpha}(\beta)=-\frac{k}{2}$

i.e. $1+\alpha\beta+\frac{\beta}{\alpha}=-\frac{k}{2}$

and $\alpha\left(\frac{1}{\alpha}\right)(\beta)=-3$

i.e. $\beta=-3$

Substituting $\beta=-3$ for x into (1):

$2(-3)^3+(-3)^2-k(-3)+6=0$

$-54+9+3k+6=0$

$3k=39$

$\therefore k=13$

7	HSC 04		

Let $P(x)=(x+1)(x-3)Q(x)+a(x+1)+b$, where $Q(x)$ is a polynomial and a and b are real numbers.

When $P(x)$ is divided by $(x+1)$ the remainder is -11.
When $P(x)$ is divided by $(x-3)$ the remainder is 1.

(i) What is the value of b? 1

(ii) What is the remainder when $P(x)$ is divided by $(x+1)(x-3)$? 2

(i) $P(x)=(x+1)(x-3)Q(x)+a(x+1)+b$

Now $P(-1)=-11$

$\therefore (-1+1)(-1-3)Q(x)+a(-1+1)+b=-11$

$0+0+b=-11$

$\therefore b=-11$

(ii) Now $P(3)=1$

$\therefore (3+1)(3-3)Q(x)+a(3+1)+b=1$

$\therefore 0+a(3+1)+b=1$

$\therefore 3a+a+b=1$

$\therefore 4a-11=1$

(substituting for b)

$\therefore 4a=12$

$\therefore a=3.$

8	HSC 06		

The cubic polynomial $P(x)=x^3+rx^2+sx+t$, where r, s and t are real numbers, has three real zeros, 1, α and $-\alpha$.

(i) Find the value of r. 1

(ii) Find the value of $s+t$. 2

(i) $P(x) = x^3+rx^2+sx+t.$

The sum of the zeros $= -\dfrac{b}{a}$

$\therefore \quad 1+\alpha+(-\alpha) = -\dfrac{r}{1}$

$\therefore \quad r=-1.$

(ii) *METHOD* 1

Since 1 is a zero of $P(x)$,

then $P(1)=0$ (Factor theorem)

$P(1) = (1)^3+r(1)^2+s(1)+t$

$0 = 1+r+s+t.$

But $r=-1$,

$0 = 1-1+s+t$

$\therefore \quad s+t=0.$

(ii) *METHOD* 2

The product of zeros $= -\dfrac{d}{a}.$

$1\times\alpha\times(-\alpha) = -\dfrac{t}{1}$

$\alpha^2 = t.$

The sum of the zeros taken 2 at a time $= \dfrac{c}{a}.$

$1\alpha+\alpha(-\alpha)+(-\alpha)1 = \dfrac{s}{1}$

$\alpha-\alpha^2-\alpha = s$

$s=-\alpha^2$

$\therefore \quad s+t=-\alpha^2+\alpha^2$

$=0.$

9	HSC 07		

The polynomial $P(x)=x^2+ax+b$ has a zero at $x=2$. When $P(x)$ is divided by $x+1$, the remainder is 18. Find the values of a and b. 3

Q2c $P(x)=x^2+ax+b$,

$P(2)=4+2a+b=0$, $\therefore 2a+b=-4$.

$P(-1)=1-a+b=18$, $\therefore a-b=-17$.

Solve simultaneously, $a=-7$ and $b=10$.

10	HSC 08		

The polynomial $p(x)$ is given by $p(x)=ax^3+16x^2+cx-120$, where a and c are constants. The three zeros of $p(x)$ are -2, 3 and α. Find the value of α. 3

Q2c $p(x)=ax^3+16x^2+cx-120$,

$p(-2)=a(-2)^3+16(-2)^2+c(-2)-120=0$,

$\therefore 4a+c=-28$ ..(1)

$p(3)=a(3)^3+16(3)^2+c(3)-120=0$,

$\therefore 9a+c=-8$ ....(2)

(2) − (1): $5a=20$,

$a=4$ and $c=-44$.

$\therefore p(x)=4x^3+16x^2-44x-120$

$=4(x^3+4x^2-11x-30)$

$(-2)(3)(\alpha)=30$, $\therefore \alpha=-5$.

11	HSC 09		

The polynomial $p(x)=x^3-ax+b$ has a remainder of 2 when divided by $(x-1)$ and a remainder of 5 when divided by $(x+2)$. Find the values of a and b. 3

Q2a $p(x)=x^3-ax+b$.

Divided by $x-1$, $R=p(1)=1-a+b=2$,

$\therefore -a+b=1$ ......(1)

Divided by $x+2$, $R=p(-2)=-8+2a+b=5$,

$\therefore 2a+b=13$ ......(2)

(2) − (1): $3a=12$,

$\therefore a=4$ and $b=5$.

12	HSC 10

Let $P(x) = (x+1)(x-3)\,Q(x) + ax + b$,

where $Q(x)$ is a polynomial and a and b are real numbers.

The polynomial $P(x)$ has a factor of $x-3$.

When $P(x)$ is divided by $x+1$ the remainder is 8.

(i) Find the values of a and b. **2**

(ii) Find the remainder when $P(x)$ is divided by $(x+1)(x-3)$. **1**

(i) $P(3) = 0 \Rightarrow 0 = 3a + b$

$P(-1) = 8 \Rightarrow 8 = -a + b$

$\therefore a = -2, b = 6.$

(ii) The remainder is $-2x+6$.

13	HSC 11

Let $P(x) = x^3 - ax^2 + x$ be a polynomial, where a is a real number. **3**

When $P(x)$ is divided by $x-3$ the remainder is 12.

Find the remainder when $P(x)$ is divided by $x+1$.

(a) $P(3) = 12, \therefore 27 - 9a + 3 = 12, \therefore a = 2$

$\therefore P(-1) = -1 - 2 - 1 = -4$

$\therefore$ Remainder is -4.

14	HSC 12

A polynomial equation has roots α, β and γ where **1**

$$\alpha + \beta + \gamma = -2, \quad \alpha\beta + \alpha\gamma + \beta\gamma = 3 \quad \text{and} \quad \alpha\beta\gamma = 1.$$

Which polynomial equation has the roots α, β and γ?

(A) $x^3 + 2x^2 + 3x + 1 = 0$ (C) $x^3 - 2x^2 + 3x + 1 = 0$

$\boxed{\text{(B)} \quad x^3 + 2x^2 + 3x - 1 = 0}$ (D) $x^3 - 2x^2 + 3x - 1 = 0$

$x^3 - \sum \alpha . x^2 + \sum \alpha\beta . x - \prod \alpha$

15	HSC 12

When the polynomial $P(x)$ is divided by $(x+1)(x-3)$, the remainder is $2x+7$. **1**

What is the remainder when $P(x)$ is divided by $x-3$?

(A) 1 (B) 7 (C) 9 $\boxed{\text{(D) } 13}$

$\boxed{P(x) = (x+1)(x-3)Q(x) + 2x + 7, P(3) = 13}$

16	HSC 13

The polynomial $P(x) = x^3 - 4x^2 - 6x + k$ has a factor $x-2$. **1**

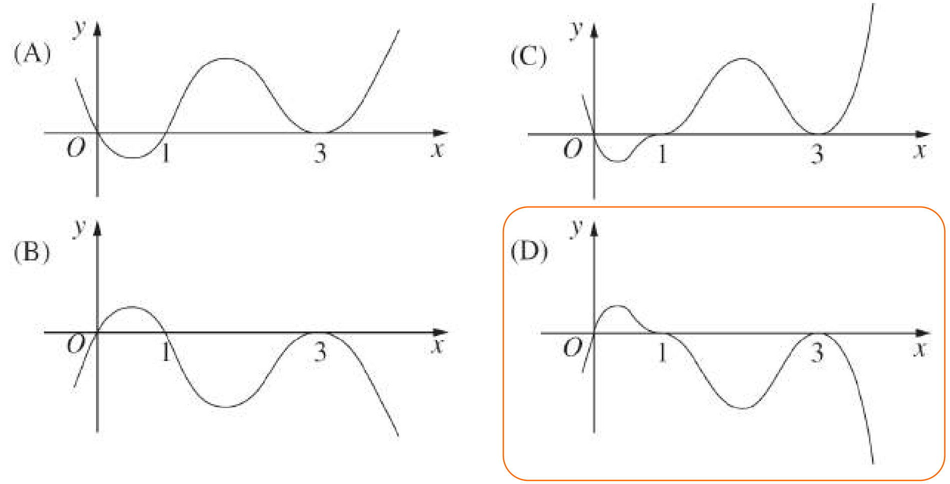
What is the value of k?

(A) 2 (B) 12 $\boxed{\text{(C) } 20}$ (D) 36

$\boxed{P(2) = 8 - 16 - 12 + k = 0, \therefore k = 20}$

17	HSC 13

Which diagram best represents the graph $y = x(1-x)^3(3-x)^2$? **1**



18	HSC 13		

The polynomial equation $2x^3 - 3x^2 - 11x + 7 = 0$ has roots α, β and γ. Find $\alpha\beta\gamma$. **1**

$\alpha\beta\gamma = -\frac{7}{2}$

19	HSC 14		

Which group of three numbers could be the roots of the polynomial equation $x^3 + ax^2 - 41x + 42 = 0$? **1**

(A) 2, 3, 7 (B) 1, −6, 7 (C) −1, −2, 21 (D) −1, −3, −14

$\prod \alpha = -42, \sum \alpha\beta = -41$

20	HSC 14		

The remainder when the polynomial $P(x) = x^4 - 8x^3 - 7x^2 + 3$ is divided by $x^2 + x$ is $ax + 3$. What is the value of a? **1**

(A) −14 (B) −11 (C) −2 (D) 5

$x^4 - 8x^3 - 7x^2 + 3 = x(x+1)Q(x) + ax + 3$

Put $x = -1, 1 + 8 - 7 + 3 = -a + 3, \therefore a = -2$

21	HSC 15		

What is the remainder when $x^3 - 6x$ is divided by $x + 3$? **1**

(A) −9 (B) 9 (C) $x^2 - 2x$ (D) $x^2 - 3x + 3$

$(-3)^3 + 18 = -9$

22	HSC 15		

Consider the polynomials $P(x) = x^3 - kx^2 + 5x + 12$ and $A(x) = x - 3$.

(i) Given that $P(x)$ is divisible by $A(x)$, show that $k = 6$. **2**

(ii) Find all the zeros of $P(x)$ when $k = 6$. **1**

(i) $P(3) = 0, \therefore 27 - 9k + 15 + 12 = 0, \therefore k = 6$

(ii) $x^3 - 6x^2 + 5x + 12 = (x-3)(x^2 - 3x - 4)$

$= (x-3)(x-4)(x+1)$

$\therefore x = 3, 4, -1.$

23	HSC 16		

What is the remainder when $2x^3 - 10x^2 + 6x + 2$ is divided by $x - 2$? **1**

(A) −66 (C) $-x^3 + 5x^2 - 3x - 1$

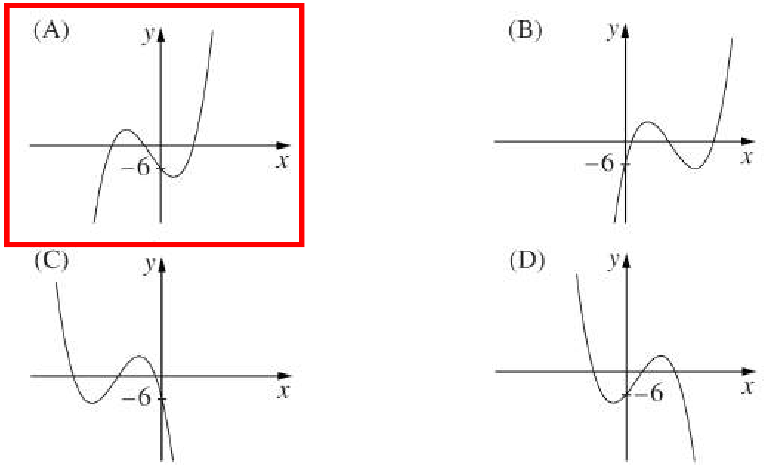
(B) −10 (D) $x^3 - 5x^2 + 3x + 1$

$\text{Remainder} = 2(2)^3 - 10(2)^2 + 6(2) + 2 = -10$

24	HSC 16		

Consider the polynomial $p(x) = ax^3 + bx^2 + cx - 6$ with a and b positive. **1**

Which graph could represent $p(x)$?



25	HSC 17		

Which polynomial is a factor of $x^3 - 5x^2 + 11x - 10$? **1**

A. $x - 2$ C. $11x - 10$

B. $x + 2$ D. $x^2 - 5x + 11$

26	HSC 18		

The polynomial $2x^3 + 6x^2 - 7x - 10$ has zeros α, β and γ. **1**

What is the value of $\alpha\beta\gamma(\alpha + \beta + \gamma)$?

A. -60 C. 15

B. -15 D. 60

27	HSC 18		

The diagram shows the graph of $y = a(x + b)(x + c)(x + d)^2$. **1**

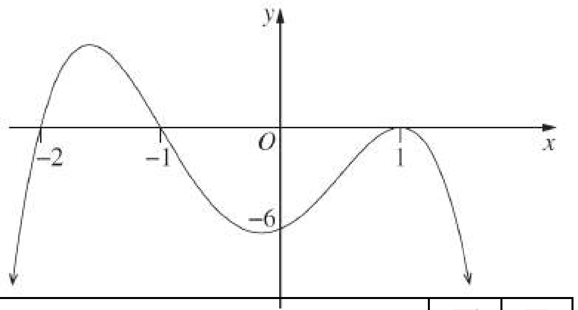
What are possible values of a, b, c and d?

A. $a = -6,\ b = -2,\ c = -1,\ d = 1$

B. $a = -6,\ b = 2,\ c = 1,\ d = -1$

C. $a = -3,\ b = -2,\ c = -1,\ d = 1$

D. $a = -3,\ b = 2,\ c = 1,\ d = -1$



28	HSC 18		

Consider the polynomial $P(x) = x^3 - 2x^2 - 5x + 6$.

(i) Show that $x = 1$ is a zero of $P(x)$. **1**

(ii) Find the other zeros. **2**

(i) ***Sample answer:***

$P(1) = 1^3 - 2(1)^2 - 5(1) + 6$

$= 1 - 2 - 5 + 6$

$= 0$

$\therefore x = 1$ is a zero of $P(x)$

(ii) ***Sample answer:***

$$\begin{array}{r} x^2 - x - 6 \\ x - 1\overline{)x^3 - 2x^2 - 5x + 6} \\ \underline{x^3 - x^2} \\ -x^2 - 5x \\ \underline{-x^2 + x} \\ -6x + 6 \\ \underline{-6x + 6} \end{array}$$

$\therefore P(x) = (x - 1)(x^2 - x - 6)$

$= (x - 1)(x + 2)(x - 3)$

other zeros are -2, 3

29	HSC 19		

Let $P(x) = qx^3 + rx^2 + rx + q$ where q and r are constants, $q \neq 0$. One of the zeros of $P(x)$ is -1. Given that α is a zero of $P(x)$, $\alpha \neq -1$, which of the following is also a zero? **1**

A. $-\dfrac{1}{\alpha}$ B. $-\dfrac{q}{\alpha}$ C. $\dfrac{1}{\alpha}$ D. $\dfrac{q}{\alpha}$

30	HSC 19		

Find the polynomial $Q(x)$ that satisfies $x^3 + 2x^2 - 3x - 7 = (x - 2)Q(x) + 3$. **2**

$$\begin{array}{r} x^2 + 4x + 5 \\ x - 2\overline{)x^3 + 2x^2 - 3x - 7} \\ \underline{x^3 - 2x^2} \\ 4x^2 - 3x \\ \underline{4x^2 - 8x} \\ 5x - 7 \\ \underline{5x - 10} \\ 3 \end{array}$$

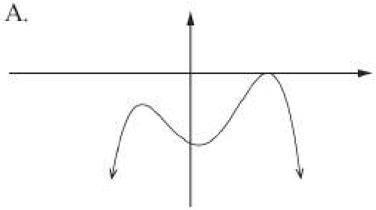
$x^3 + 2x^2 - 3x - 7 = (x - 2)(x^2 + 4x + 5) + 3$

$\therefore Q(x) = x^2 + 4x + 5$

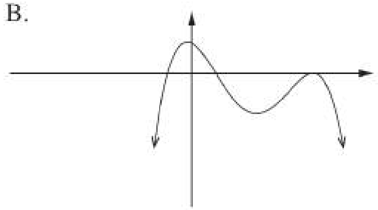
31	HSC 20		

A monic polynomial $p(x)$ of degree 4 has one repeated zero of multiplicity 2 and is divisible by $x^2 + x + 1$. Which of the following could be the graph of $p(x)$? **1**

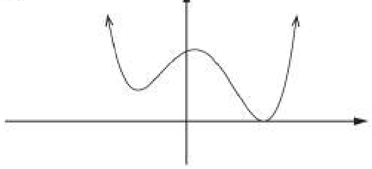
A.



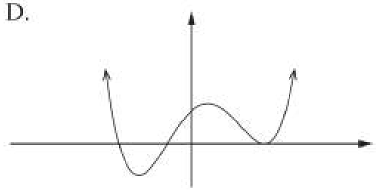
B.



C.



D.



32	HSC 20	☑	☒

Let $P(x) = x^3 + 3x^2 - 13x + 6$.

(i) Show that $P(2) = 0$. **1**

(ii) Hence, factor the polynomial $P(x)$ as $A(x)B(x)$, where $B(x)$ is a quadratic polynomial. **2**

(i) $P(2) = 2^3 + 3(2)^2 - 13(2) + 6 = 0$

(ii)

$$\begin{array}{r|l} & x^2 + 5x - 3 \\ \hline x - 2 & x^3 + 3x^2 - 13x + 6 \\ & \underline{x^3 - 2x^2} \\ & 5x^2 - 13x \\ & \underline{5x^2 - 10x} \\ & -3x + 6 \\ & \underline{-3x + 6} \\ & 0 \end{array}$$

$\therefore P(x) = (x-2)(x^2 + 5x - 3)$

33	HSC 21	☑	☒

What is the remainder when $P(x) = -x^3 - 2x^2 - 3x + 8$ is divided by $x + 2$? **1**

A. -14 B. -2 C. 2 D. 14

34	HSC 21	☑	☒

The roots of $x^4 - 3x + 6 = 0$ are α, β, γ and δ. **1**

What is the value of $\dfrac{1}{\alpha} + \dfrac{1}{\beta} + \dfrac{1}{\gamma} + \dfrac{1}{\delta}$?

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta} = \frac{\beta\gamma\delta + \alpha\gamma\delta + \alpha\beta\delta + \alpha\beta\gamma}{\alpha\beta\gamma\delta}$$

$x^4 - 3x + 6 = 0$ $\quad\therefore a = 1, b = 0, c = 0, d = -3, e = 6$

$\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\delta\gamma = \dfrac{-d}{a}$ (coeff of x)

$= 3$

and $\alpha\beta\gamma\delta = \dfrac{e}{a}$ (constant term) Therefore $\dfrac{1}{\alpha} + \dfrac{1}{\beta} + \dfrac{1}{\gamma} + \dfrac{1}{\delta} = \dfrac{3}{6} = \dfrac{1}{2}$.

$= 6$

35	HSC 22	☑	☒

Let $P(x)$ be a polynomial of degree 5. When $P(x)$ is divided by the polynomial $Q(x)$, the remainder is $2x + 5$. Which of the following is true about the degree of Q? **1**

A. The degree must be 1.

B. The degree could be 1.

C. The degree must be 2.

D. The degree could be 2.

36	HSC 22	☑	☒

The monic polynomial, P, has degree 3 and roots α, β, γ. **3**

It is given that $\alpha^2 + \beta^2 + \gamma^2 = 85$ and $P'(\alpha) + P'(\beta) + P'(\gamma) = 87$.

Find $\alpha\beta + \beta\gamma + \gamma\alpha$.

$\alpha^2 + \beta^2 + \gamma^2 = 85$

$P'(\alpha) + P'(\beta) + P'(\gamma) = 87$

Let $P(x) = x^3 + ax^2 + bx + c$

Then $a = -(\alpha + \beta + \gamma)$

$b = \alpha\beta + \beta\gamma + \gamma\alpha$

$c = -\alpha\beta\gamma$

$P'(x) = 3x^2 + 2ax + b$

$$\begin{aligned} &P'(\alpha) + P'(\beta) + P'(\gamma) \\ &= 3(\alpha^2 + \beta^2 + \gamma^2) + 2a(\alpha + \beta + \gamma) + 3b \\ &= 3(\alpha^2 + \beta^2 + \gamma^2) - 2(\alpha + \beta + \gamma)^2 + 3(\alpha\beta + \beta\gamma + \gamma\alpha) \\ &= 3(\alpha^2 + \beta^2 + \gamma^2) - 2(\alpha^2 + \beta^2 + \gamma^2) - 4(\alpha\beta + \beta\gamma + \gamma\alpha) + 3(\alpha\beta + \beta\gamma + \gamma\alpha) \\ &= \alpha^2 + \beta^2 + \gamma^2 - (\alpha\beta + \beta\gamma + \gamma\alpha) \end{aligned}$$

$87 = 85 - (\alpha\beta + \beta\gamma + \gamma\alpha)$

$\alpha\beta + \beta\gamma + \gamma\alpha = 85 - 87$

$= -2$